

CRYPTOGRAPHY AND NETWORK SECURITY

IV B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY06	PCC	L	T	P	C	CIA	SEE	Total
		3	-	-	3	40	60	100
Contact Classes:54	Tutorial Classes:	Practical Classes:			Total Classes:54			
COURSE OBJECTIVES The course will enable the students to: 1. To understand the objectives of information security. 2. To inculcate confidentiality, integrity, authentication and availability. 3. To Introduce various cryptographic algorithms and public-key cryptosystem 4. Demonstrate a PGP key pair and encrypted e- mail message. 5. To give a clear idea on fundamental ideas of public-key cryptography, web security and Firewalls.								
COURSE OUTCOMES At the end of the course, student will be able to: 1. Ability to understand basic cryptographic algorithms. 2. Ability to understand message and web authentication and security issues. 3. Ability to identify information system requirements for both of them such as client and server. 4. Understand the current legal issues towards information security. 5. Understand Intrusions and intrusion detection.								
UNIT-I	CRYPTOGRAPHY CONCEPTS					CLASSES :12		
What is Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security. Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks.								
UNIT-II	SYMMETRIC KEY CIPHERS AND ASYMMETRIC KEY CIPHERS					CLASSES:12		
Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, and Knapsack Algorithm.								
UNIT-III	CRYPTOGRAPHIC HASH FUNCTIONS					CLASSES:14		
Message Authentication, Secure Hash Algorithm (SHA-512), Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme. Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure.								

UNIT-IV	TRANSPORT-LEVEL SECURITY	CLASSES:10
Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH). Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security.		
UNIT-V	E-MAIL SECURITY	CLASSES:10
Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange.		
TEXT BOOKS		
1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition. 2. Cryptography and Network Security: Atul Kahate, McGraw Hill, 3rd Edition.		
REFERENCE BOOKS		
1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition. 2. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3rd Edition. 3. Information Security, Principles, and Practice: Mark Stamp, Wiley India. 4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH. 5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning. 6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.		